

ID6002W : RL Course Project Report

Industrial Inventory Control

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1 Problem and Assigned Configuration

The industrial inventory control problem requires an agent to manage order quantities across three distinct products over discrete time steps to minimize total operational costs. The cost function penalizes holding excess stock, placing orders, discarding expired or overstocked units, and experiencing stockouts when demand exceeds available inventory. Stochastic customer demand and non-zero lead times create delayed arrivals, requiring policies to coordinate multi-echelon pipelines and capacity constraints.

- **Project version:** IITM-6002W-RL-Inventory-2026-v1
- **Roll number:** DA25M622
- **Variant ID:** V022
- **Config fingerprint:** 7d77fc79debf206a
- **Demand multiplier profile:** [1.0, 1.1, 0.9]
- **Initial inventory profile:** [110, 100, 90]
- **Lead-time delay profile:** [0.1, 0.0, 0.05]

2 Techniques Attempted

Five distinct reinforcement learning techniques were implemented and evaluated: Advantage Actor-Critic (A2C), Proximal Policy Optimization (PPO), Deep Q-Network (DQN), Double Deep Q-Network (Double DQN), and Monte Carlo Policy Gradient (REINFORCE with Baseline).

Technique	Observations Used	Key Design Choices
A2C	Dict observation: <code>inventory</code> (3), <code>arrival_pipeline</code> (12), <code>demand_history</code> (21), <code>day</code> (1), <code>capacity_utilisation</code> (1)	Synchronous multi-step actor-critic using Stable-Baselines3 <code>MultiInputPolicy</code> , parallelized over 4 vectorized environments, shared feature extractor with separate policy and value heads.
PPO	Same dictionary state representation (total 38 flattened scalar features)	Clipped surrogate objective with GAE; <code>MultiInputPolicy</code> ; 4 parallel workers; mini-batch updates with value function and entropy regularization terms.
DQN	Flattened dictionary state vector mapped into a single observation array (38 features)	Discrete action space flattened from <code>MultiDiscrete([11, 11, 11])</code> to <code>Discrete(1331)</code> via action wrapper; experience replay buffer size 100,000; ϵ -greedy exploration decay; target Q-network.
Double DQN	Flattened state vector (38 features)	Same <code>Discrete(1331)</code> action wrapper as DQN; decoupled action selection via online Q-network and action evaluation via target Q-network to eliminate Q-value overestimation bias.
REINFORCE with Baseline	Dict converted to tensor: <code>inventory</code> , <code>arrival_pipeline</code> , <code>demand_history</code> , <code>day</code> , <code>capacity_utilisation</code>	Custom PyTorch policy and value networks; 3-head categorical action output (11×3 logits); parameterized baseline value network trained via MSE loss; normalized returns and gradient clipping.

Table 1: Summary of techniques attempted.

3 Training and Experimental Setup

All models were evaluated under an identical protocol using a deterministic evaluation over 20 evaluation episodes, with the reward defined by the environment as $r_t = -\text{cost}_t/100$. Random seed 2026 was fixed across environment instances, data generators, and network initializations. Hyperparameters for all five models were systematically tuned using Optuna’s Tree-structured Parzen Estimator (TPE) sampler over 15 optimization trials each.

- **Training steps/episodes:**

- A2C: 200,000 timesteps (final run) across 4 parallel environments.
- PPO: 200,000 timesteps (final run) across 4 parallel environments.
- DQN: 100,000 timesteps with 1,000 learning start steps.
- Double DQN: 80,000 timesteps with 500 learning start steps.
- REINFORCE with Baseline: 500 complete rollout episodes (1,000 max steps per episode).

- **Discount factor and learning rates:**

- A2C: $\gamma = 0.995$, learning rate = 1.04×10^{-4} , $\lambda_{\text{GAE}} = 0.95$.
- PPO: $\gamma = 0.99$, learning rate = 2.50×10^{-4} , $\lambda_{\text{GAE}} = 0.90$, clip range = 0.3.
- DQN: $\gamma = 0.99$, learning rate = 7.00×10^{-4} , batch size = 64.
- Double DQN: $\gamma = 0.99$, learning rate = 5.00×10^{-4} , batch size = 128.
- REINFORCE: $\gamma = 0.99$, actor learning rate = 3.00×10^{-4} , critic learning rate = 5.00×10^{-4} .

- **Exploration strategy:**

- DQN: ϵ -greedy schedule decaying over an exploration fraction of 0.30 to a final $\epsilon = 0.05$.
- Double DQN: ϵ -greedy schedule decaying over an exploration fraction of 0.25 to a final $\epsilon = 0.10$.

- A2C, PPO, and REINFORCE: Stochastic categorical sampling during training parameterized by action logits, assisted by entropy regularization (coefficients: 2.00×10^{-4} for A2C, 0.01 for PPO, and 0.01 for REINFORCE).
- **Reward shaping:** No artificial reward shaping was introduced; the environment default reward ($-\text{total_cost}/100$) was preserved to ensure uncorrupted policy convergence to the true cost objective. Advantage normalization was applied to policy gradient baselines.
- **Random seeds:** Seed 2026 was used as the universal master seed across training environments, policy network initialization, and evaluation seeds (`SEED + episode_idx` for custom rollouts).

4 Results

Validation performance was measured across 20 deterministic evaluation episodes locally prior to leaderboard submission. Table 2 reports local evaluation performance alongside confirmed public leaderboard submission outcomes.

Technique	Local Mean Cost	Public Leaderboard Cost	Submission ID	Network Arch
PPO	81,770.00	112,479.75	3418	MLP
Double DQN	104,805.00	134,677.50	3510	MLP
DQN	108,775.00	116,214.38	3551	MLP
A2C	153,264.25	154,022.75	3666	MLP
REINFORCE w/ Baseline	1,591,876.80	1,290,798.88	3621	MLP

Table 2: Validation and public leaderboard performance across attempted techniques.

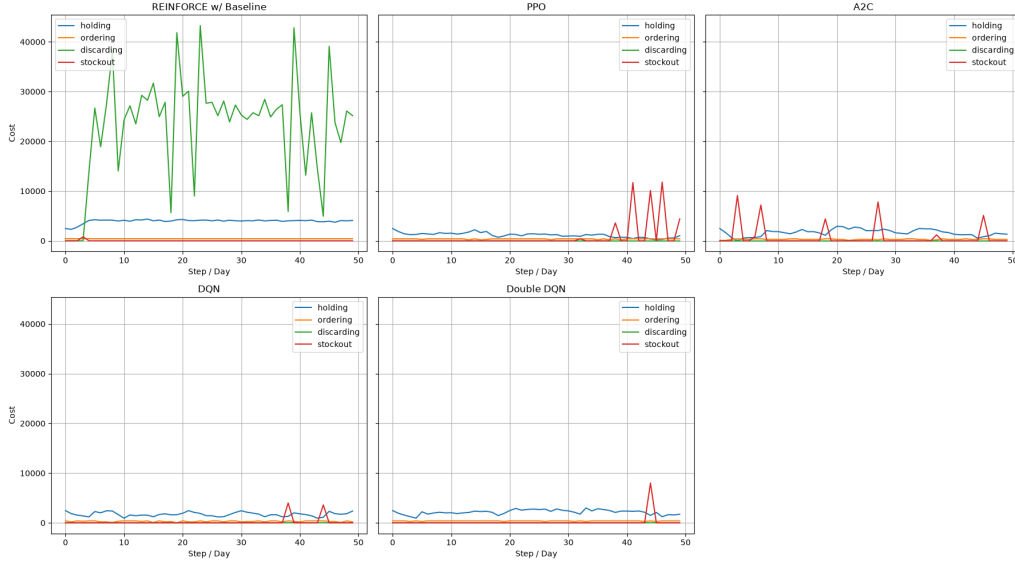


Figure 1: Breakdown of cost components (holding, ordering, discarding, stockout) across time steps in an evaluation episode for all five models.

5 Analysis and Observations

- **Comparative Performance:** PPO achieved the lowest overall cost (local mean of 81,770.00 and public score of 112,479.75). The clipped surrogate objective prevented destructive gradient updates, allowing steady policy refinement over multi-product action interactions.

- **Discrete Action Space Complexity:** Both DQN (local: 108,775.00) and Double DQN (local: 104,805.00) handled the $11^3 = 1331$ action space effectively, but the combinatorially large action space slowed convergence and increased sensitivity to target network update intervals (1000 to 2000 steps). Double DQN reduced local mean cost compared to standard DQN by suppressing overestimation bias.
- **Value of Arrival Pipeline and State Features:** The observation vector consisted of 38 dimensions: current inventory (3), in-transit arrival pipeline (12), demand history (21), current day (1), and capacity utilization (1). Policies that successfully suppressed stockout and discard costs relied heavily on the arrival pipeline and demand history to anticipate deliveries and hedge against non-zero lead-time delays (up to 10% delay probability for Product 0).
- **Failure Modes and Cost Profile Analysis:** As evidenced in Figure 1, the primary failure mode for weaker policies (such as REINFORCE with Baseline) was severe stockout spikes and high discard costs stemming from erratic ordering swings. A2C maintained relatively uniform ordering but incurred sustained holding costs due to over-buffering inventory. PPO established a tighter balance, maintaining low holding costs while rarely triggering stockouts or discards.
- **Sample Efficiency and High Variance:** REINFORCE with Baseline exhibited catastrophic variance and sample inefficiency, averaging costs above 1.2×10^6 . Because Monte Carlo returns require full-trajectory rollouts without bootstrapping, policy updates were delayed and noisy compared to temporal difference and actor-critic methods.

6 Key Learning

1. **Policy Gradient Stability with Clipping:** PPO’s clipped objective and GAE advantage estimation significantly outperformed unclipped actor-critic (A2C) and Monte Carlo policy gradients (REINFORCE), proving essential for high-dimensional multi-product inventory balance.
2. **Scalability Limits of Value-Based Methods on Joint Spaces:** Factoring a MultiDiscrete action space into a flat 1331-dimensional discrete space allows DQN/Double DQN to learn valid policies, but introduces severe sample complexity and high replay buffer requirements compared to multi-head actor policies.
3. **Importance of Arrival Pipeline Tracking in Delayed MDPs:** In inventory environments with non-zero lead times, observing current on-hand stock alone is insufficient; pipeline tracking is critical to prevent redundant over-ordering and subsequent discard penalties.
4. **Hyperparameter Sensitivity in RL Control:** Systematically tuning discount factor γ close to 0.99–0.995 and applying entropy regularization was decisive in preventing premature convergence to sub-optimal policies like zero-ordering or excessive hoarding.

References

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